Cofactor-pencil reconstruction: recovery and rejection tests

November 13, 2026. Unsigned producer evidence [pcc:unverified].

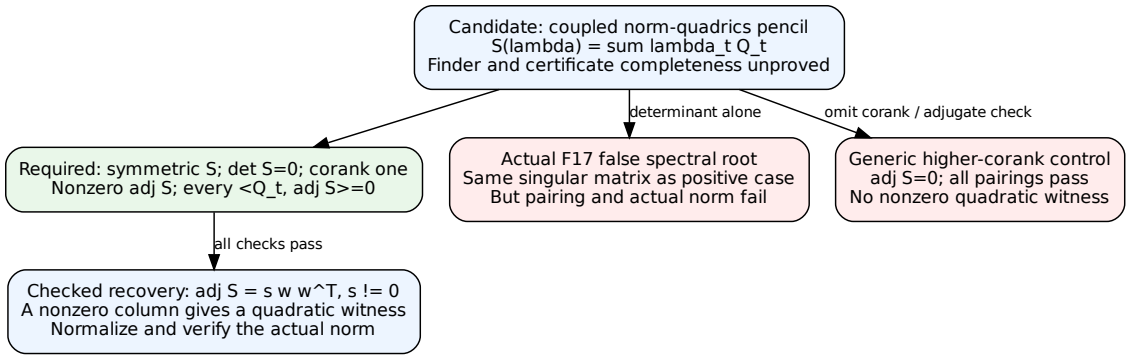
Candidate. Encode the norm coefficient equations as homogeneous quadrics $q_t(w) = \sum_{i,j} Q_t(i,j) w_i w_j$ after eliminating the target branch. Search a small symmetric pencil $S(\lambda) = \sum_t \lambda_t Q_t$ for a corank-one singular point whose cofactor pairings all vanish. The higher-dimensional norm-to-pencil bridge is not yet formalized; certificate completeness and an efficient finder remain unproved. This stage checks sufficient recovery and exact controls.

Checked recovery. Over any field, let a supplied symmetric $m \times m$ matrix S satisfy $\det S = 0$, $\operatorname{rank} S + 1 = m$, and $\operatorname{adj} S \neq 0$. If

$$\langle Q_t, \operatorname{adj} S \rangle := \sum_{i,j} Q_t(i,j)(\operatorname{adj} S)_{ij} = 0 \quad \text{for every } t,$$

then a nonzero vector w satisfying every $q_t(w) = 0$ exists. Lean proves the adjugate rank bound, all minor identities and scaled outer-product extraction. The generic theorem does not assume S belongs to a particular pencil.

Indeed $S \operatorname{adj} S = 0$ and the matrix rank inequality force $\operatorname{rank}(\operatorname{adj} S) \leq 1$. Symmetry and the verified nonzero condition give $\operatorname{adj} S = sww^T$, $s \neq 0$. Every pairing becomes $sq_t(w) = 0$. No square root is needed.



Actual norm fixture, false positives and evidence

Two-support pencil. With $w = (v, u)$ after branch elimination,

$$Q_1 = \begin{pmatrix} 1 & 0 \\ 0 & -U \end{pmatrix}, \quad Q_2 = \begin{pmatrix} 0 & y \\ y & -V \end{pmatrix}, \quad S(a,b) = \begin{pmatrix} a & by \\ by & -aU - bV \end{pmatrix}.$$

Here $U = q + x_1 + x_2$ and $V = q^2 + \alpha + qU - x_1 x_2$ for curve coefficient α. Lean checks

$$\det S = -Ua^2 - Vab - y^2 b^2, \quad \langle Q_1, \operatorname{adj} S \rangle = -2Ua - Vb, \quad \langle Q_2, \operatorname{adj} S \rangle = -Va - 2y^2 b.$$

Positive and negative actual-curve controls. On $y^2 = x^3 + 2x + 2$ over $\mathbb{F}_{17}$ with supports $(5,1), (6,3)$, use $(a,b) = (3,1)$. Target $(10,6)$ gives $(U,V) = (4,10)$; target $(0,6)$ gives $(11,6)$. Both give exactly $S = \left(\begin{smallmatrix} 3 & 6 \\ 6 & 12 \end{smallmatrix}\right)$, with determinant zero and nonzero adjugate. Its normalized adjugate column has slope15. Both cofactor tests and the actual positive norm pass. For the negative target a cofactor test and the actual recovered norm fail. Thus a determinant root and nonzero adjugate alone are insufficient.

Higher-corank control. In the coordinate-square pencil, $S = \operatorname{diag}(1,0,0)$ is nonzero singular with zero adjugate. Every cofactor pairing passes, although $w_i^2 = 0$ for all i has no nonzero witness. This is a generic control, not a curve norm instance.

Formal scope. Main theorem: `NormDualPencil.recoverQuadratic`. Exact formulas and all control statements are mapped in the README. Source `71f1bfb22`; build005 passed3147 jobs; tracked-source gate passed62 declarations with zero additional trusted axioms. Earlier declaration hashes are unchanged. Import-inclusive fresh status is in the evidence directory. Failed builds and final source hashes are retained; all standing is unsigned.

Native determinant record `IDC1h25bf8a5444f4976f` and cofactor record `IDC1h76b1a590e132b252` each have32 statement-derived and32 extra-stream points and a source hash. They certify expression checks only, not Lean execution or a search bound.

Open obligations. A general norm-to-pencil completeness/regularity proof and a constructive finder costing $2^{o(n)}$ bit operations remain absent, with $n = \lceil \log_2 r \rceil$ for prime subgroup order r. Polynomial determinant evaluation does not find the required cofactor point. No empirical scaling run, paid compute or performance claim was made. Sources: pinned Mathlib adjugate/rank declarations and actual norm proofs; previous kernel/jet scope reviewed. KB unavailable; no novelty asserted.